
Macro-Informed Machine-Learning ROE Forecasts and Residual Income Valuation: Evidence from U.S. Commercial Banks

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ABSTRACT

The residual income valuation (RIV) model is well suited to banks, but standard implementations project abnormal profitability with a single fixed persistence parameter that does not respond to changing macro-financial conditions. The bank-profitability literature documents pronounced cyclicalities, and the machine-learning literature documents gains from nonlinear forecasting, yet these strands have seldom been connected to accounting-based valuation. This study examines whether macro-informed machine-learning forecasts of return on equity (ROE) improve RIV accuracy for large U.S. commercial banks. In the first stage, three econometric panel models and nine machine-learning models forecast one-quarter-ahead ROE for 20 banks (680 bank-quarters, 2017Q1–2025Q4) under a common chronological train–test design. LightGBM produces the strongest point-estimate performance (test RMSE = 3.308; test R^2 = 0.405, against 4.399 for the Pooled OLS benchmark), but Diebold–Mariano tests do not establish statistically significant superiority over the main benchmark models. In the second stage, a valuation-specific LightGBM architecture—direct forecasts for quarters 1–4, a recursive continuation for quarters 5–8 and capped persistence decay thereafter—is embedded in a clean-surplus-consistent RIV framework and applied to JPMorgan Chase & Co. at nine rolling valuation dates (2024Q1–2026Q1). Relative to the traditional persistence-based RIV, MAPE falls from 44.07% to 21.91% and RMSE from \$121.25 to \$70.16. The study contributes a like-for-like forecasting benchmark and an auditable integration of machine-learning forecasts into accounting-based valuation. The valuation evidence is specific to one firm and period, and a sizeable gap to market prices persists during 2025.

1. Introduction

Equity valuation underpins investment decisions, capital allocation and regulatory oversight. Among available approaches, the Residual Income Valuation (RIV) model offers an accounting-based alternative to discounted cash flow analysis. Rather than forecasting an uncertain stream of free cash flows, RIV anchors value to current book value plus the present value of expected residual income—earnings in excess of the return shareholders require on invested capital (Ohlson, 1995). The model rests on the clean surplus relation linking changes in book value to earnings and dividends (Peasnell, 1982; Feltham & Ohlson, 1995).

RIV is particularly suited to banks. Because bank balance sheets are dominated by financial assets, free cash flow is conceptually ambiguous and empirically unstable (Penman & Sougiannis, 1998; Dermine, 2010). RIV instead works from earnings and book value, both audited and comparable across institutions, and residual income models have been found to produce lower valuation errors than dividend discount or free cash flow models (Penman & Sougiannis, 1998; Frankel & Lee, 1998; Dechow, Hutton, & Sloan, 1999; Jiang & Lee, 2005).

The model’s central limitation lies in the treatment of abnormal profitability. Standard implementations assume that the spread between return on equity (ROE) and the cost of equity decays at a fixed rate estimated from historical data (Ohlson, 1995; Dechow et al., 1999). This assumption may be restrictive for banks, whose profitability varies with monetary policy, credit conditions and bank-specific fundamentals

(Demirgüç-Kunt & Huizinga, 1999; Athanasoglou, Brissimis, & Delis, 2008; Albertazzi & Gambacorta, 2009). A fixed persistence parameter is not designed to represent such state-dependent dynamics explicitly, and Dechow et al. (1999) show that the persistence assumption has a first-order effect on valuation accuracy.

Machine-learning methods offer one way to relax this restriction. Tree-based ensembles can represent nonlinear relationships and interactions among bank fundamentals and macro-financial conditions without prespecifying their functional form (Gu, Kelly, & Xiu, 2020). However, evidence that a model forecasts ROE well does not imply that its forecasts improve valuation: valuation accuracy also depends on how forecast errors interact with discounting and terminal-value mechanics (Penman & Sougiannis, 1998). Forecasting and valuation are therefore related but distinct empirical questions.

The existing literature appears to provide limited evidence connecting these elements. RIV applications typically retain a single persistence parameter; the bank-profitability literature documents cyclicalities but rarely embeds it in a formal valuation model; and applications of gradient boosting in banking concentrate on classification problems such as insolvency and distress prediction (Climent, Momparler, & Carmona, 2019; Petropoulos, Siakoulis, Stavroulakis, & Vlachogiannakis, 2020). The closest related study, Liu and Hwang (2025), incorporates a business-cycle indicator into the Ohlson framework for Taiwanese equities, but treats the cycle as a modifier of the price–accounting relationship rather than as an input to forecasting future ROE.

This study addresses that gap with a two-stage design. The first stage compares three econometric panel models and nine machine-learning models in forecasting one-quarter-ahead ROE for 20 large U.S. commercial banks under a common chronological train–test protocol spanning the COVID-19 shock and the 2022 tightening cycle. The second stage embeds a valuation-specific LightGBM forecasting architecture in a clean-surplus-consistent RIV framework and compares its valuation accuracy with that of a traditional persistence-based RIV model for JPMorgan Chase & Co. across nine rolling valuation dates from 2024Q1 to 2026Q1. Two research questions guide the analysis. RQ1 asks which forecasting approach—econometric panel models or machine-learning algorithms—produces the most accurate out-of-sample forecasts of bank ROE. RQ2 asks whether embedding machine-learning-based ROE forecasts within the RIV framework produces intrinsic value estimates closer to observed market prices than the traditional model.

The results are mixed in an informative way. LightGBM records the lowest test root mean squared error (RMSE; 3.308) and highest test R^2 (0.405) of the models evaluated, improving on the Pooled ordinary least squares (OLS) benchmark by approximately 25% in point-estimate terms. However, none of LightGBM's pairwise Diebold–Mariano comparisons with the benchmark models is statistically significant at the 5% level. In the valuation stage, the Enhanced RIV reduces the mean absolute percentage error (MAPE) from 44.07% to 21.91% and RMSE from \$121.25 to \$70.16 relative to the traditional model, although both models remain below the market price at most valuation dates and the gap widens during the 2025 price appreciation.

The study makes three contributions. First, it provides a like-for-like comparison of econometric and machine-learning approaches to bank ROE forecasting, using one dataset, one chronological hold-out period and identical evaluation criteria, while distinguishing point-estimate performance from statistically significant predictive superiority. Second, it integrates a machine-learning forecasting architecture into an Ohlson-style RIV framework that preserves clean-surplus consistency and imposes a capped persistence mechanism on the long horizon, so that every valuation component remains auditable. Third, it evaluates valuation accuracy directly rather than inferring it from forecasting accuracy, and supports model

interpretation with native and permutation-based feature importance.

The remainder of the paper is organised as follows. Section 2 reviews the literature and develops the hypotheses. Section 3 describes the data and empirical design. Section 4 reports the empirical results. Section 5 discusses the findings, Section 6 addresses robustness and limitations, and Section 7 concludes.

2. Literature review and hypothesis development

2.1. Residual income valuation

RIV originates in [Edwards and Bell \(1961\)](#) and [Peasnell \(1982\)](#), who connected economic value to accounting numbers. [Ohlson \(1995\)](#) gave the model its modern form, in which intrinsic value equals current book value plus the present value of expected future residual income under clean surplus accounting; [Feltham and Ohlson \(1995\)](#) further separated operating and financial activities. [Penman and Sougiannis \(1998\)](#) show that RIV produces smaller valuation errors than dividend discount and free cash flow models, and [Frankel and Lee \(1998\)](#) find that the RIV value-to-price ratio predicts returns better than book-to-price alone. [Jiang and Lee \(2005\)](#) attribute RIV's advantage over dividend discounting to its joint use of earnings and book value. [Dechow et al. \(1999\)](#) confirm the model's general reliability but show that the rate at which abnormal earnings revert—the persistence parameter—has a first-order effect on accuracy. [Heinrichs, Hess, Homburg, Lorenz, and Sievers \(2013\)](#) find that the model remains reliable when several simplifying assumptions are relaxed, and [Kuo \(2016\)](#) shows that vector error correction (VECM) and vector autoregressive (VAR) specifications outperform OLS out of sample, a precedent for benchmarking several techniques rather than assuming one.

2.2. RIV and bank valuation

[Dermine \(2010\)](#) argues that bank valuation requires a model compatible with a balance sheet composed mainly of financial instruments, for which free cash flow is difficult to define consistently. [Aggelopoulos \(2017\)](#) finds that residual income approaches outperform equity cash flow approaches in bank applications. Because bank capital is closely supervised—including minimum Common Equity Tier 1 requirements under Basel III—book value of equity is closely monitored and comparable across institutions, which further supports the use of RIV in this setting.

2.3. Profitability persistence and macroeconomic cycles

The standard RIV implementation forecasts abnormal earnings with a single persistence parameter estimated from historical autocorrelation ([Ohlson, 1995](#); [Dechow et al., 1999](#)), implicitly assuming a constant decay rate regardless of the economic environment. A substantial literature questions this for banks. [Demirgüç-Kunt and Huizinga \(1999\)](#) find, across 80 countries, that macroeconomic conditions affect bank margins after controlling for bank characteristics. [Athanasoglou et al. \(2008\)](#) identify GDP growth as a driver of bank profitability through the credit-demand channel, and [Albertazzi and Gambacorta \(2009\)](#) show that GDP growth, credit expansion and the slope of the yield curve affect profitability through distinct channels. [Borio, Gambacorta, and Hofmann \(2017\)](#) find that the level and slope of the yield curve matter most when interest rates are low, and [Claessens, Kose, and Terrones \(2012\)](#) show that recessions accompanied by housing downturns are deeper and more persistent. This literature documents state-dependent profitability dynamics that a fixed persistence parameter is not designed to represent.

[Liu and Hwang \(2025\)](#) add a business-cycle indicator to the [Ohlson](#) framework using Taiwanese equity data for 1990–2019, decomposing earnings and book value into financial-statement components following [Lewellen and Resutek \(2019\)](#). They find the business-cycle indicators significant and report that VECM

and VAR specifications outperform OLS out of sample. Their design treats the business cycle as a variable modifying the cross-sectional price–accounting relationship rather than as an input to forecasting future ROE within the valuation model; the present study takes the latter approach.

2.4. Machine learning for financial forecasting

Gu et al. (2020) find that tree-based ensembles outperform linear factor models in forecasting cross-sectional stock returns, with gradient boosting among the strongest methods. Reviewing this literature, Goodell, Kumar, Lim, and Pattnaik (2021) note that accounting-based valuation remains comparatively underexplored relative to price prediction and credit scoring. Gradient boosting was formalised by Friedman (2001), and subsequent implementations—including XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017)—improved computational efficiency and regularisation for tabular prediction problems. In banking, Petropoulos et al. (2020) show that machine learning improves insolvency prediction relative to logit-based approaches, and Climent et al. (2019) find that XGBoost improves the anticipation of Euro-zone bank distress. These are classification tasks rather than continuous profitability forecasts embedded in a valuation framework.

2.5. Explainable machine learning

A common objection to machine learning in financial decision-making is limited interpretability, which matters where decisions must be explained to supervisors and auditors. Lundberg and Lee (2017) propose SHAP (SHapley Additive exPlanations), a model-agnostic attribution framework grounded in cooperative game theory in which feature contributions sum to the difference between a prediction and a baseline value. Bracke, Datta, Jung, and Sen (2019) apply SHAP to credit default predictions, and Bussmann, Giudici, Marinelli, and Papenbrock (2021) extend explainable machine learning to credit-risk management. In this literature, attributions describe predictive contributions within a fitted model; they do not identify causal effects.

2.6. Forecasting accuracy, the cost of equity and valuation accuracy

Forecasting performance depends on feature construction. The bank-profitability literature motivates combining macroeconomic and credit-cycle variables with bank fundamentals and the lagged history of the target (Albertazzi & Gambacorta, 2009; Claessens et al., 2012), with look-ahead bias avoided by construction in line with established forecast-evaluation practice (Diebold & Mariano, 1995). Penman and Sougiannis (1998) show that forecast errors in accounting variables propagate into valuation errors, but a more accurate ROE forecast does not automatically yield a more accurate intrinsic value, because valuation accuracy also depends on discounting and terminal-value mechanics.

The discount rate in RIV is the cost of equity, commonly estimated with the capital asset pricing model (CAPM; Sharpe, 1964; Lintner, 1965). Fama and French (1992, 1993) show that a single market factor leaves much of the cross-section of expected returns unexplained; the single-factor CAPM is nonetheless retained here, consistent with much of the RIV literature, for transparency and comparability between models. Because RIV's validity depends on clean surplus holding throughout the forecast horizon (Peasnell, 1982; Feltham & Ohlson, 1995), book value is best projected from forecast earnings and payout rather than forecast independently.

2.7. Research gap

Three literatures have developed largely in parallel. RIV applications retain a fixed persistence parameter even though it is a first-order driver of valuation accuracy (Dechow et al., 1999). The bank-profitability literature shows that profitability is cyclical and state-dependent, yet this is rarely operationalised as a forecasting input within a formal valuation model; even Liu and Hwang (2025) use cyclical information as a cross-sectional modifier. The machine-learning literature reports strong performance in finance and banking but concentrates on price prediction and classification. The existing literature appears to provide limited evidence on the combination of (i) a systematic benchmark of ROE forecasting models, (ii) the integration of the selected forecasting approach into a clean-surplus-consistent RIV framework, and (iii) an explicit comparison of valuation accuracy—rather than forecasting accuracy alone—against a traditional persistence-based benchmark.

2.8. Hypothesis development

The literature on nonlinear, state-dependent bank profitability suggests that flexible models may forecast ROE more accurately than linear panel specifications. This motivates the first hypothesis, evaluated with out-of-sample error metrics and pairwise Diebold–Mariano tests at the 5% significance level:

H1₀: Machine-learning forecasting models, including gradient boosting models, do not produce significantly more accurate out-of-sample one-quarter-ahead forecasts of bank ROE than traditional econometric panel models.

H1_A: Machine-learning forecasting models produce lower out-of-sample one-quarter-ahead ROE forecast errors than traditional econometric panel models for the large U.S. commercial banks examined.

Where a Diebold–Mariano test yields $p < 0.05$, the null of equal predictive accuracy is rejected, and the sign of the loss differential determines whether the difference favours the machine-learning model.

If a more flexible forecast better represents the path of abnormal profitability, it should also reduce valuation error when the remaining RIV architecture is held constant:

H2₀: The Enhanced RIV framework does not produce lower valuation error against observed market prices than the traditional persistence-based RIV model.

H2_A: The Enhanced RIV framework produces lower valuation error against observed market prices than the traditional persistence-based RIV model.

H2 is evaluated by comparing MAE, RMSE and MAPE across the nine rolling valuation dates; support requires lower values on all three metrics. Because this stage uses a small number of valuation dates and no separate formal significance test, conclusions are expressed as comparative empirical support rather than statistical rejection of the null.

3. Data and empirical design

3.1. Sample and data sources

Bank-level financial data are obtained from LSEG Workspace (Refinitiv Eikon; LSEG, 2026), which reports 29 financial-statement variables for 21 publicly listed U.S. commercial banks on a standardised quarterly template from September 2016 to March 2026. Macroeconomic data are obtained from FRED (Federal Reserve Economic Data; Federal Reserve Bank of St. Louis, 2026): real GDP growth, inflation,

unemployment, the Federal Funds rate, the CBOE Volatility Index (VIX), the 10-year/2-year Treasury yield spread, and two Senior Loan Officer Opinion Survey (SLOOS) series covering lending standards and loan demand. All eight macroeconomic series contain no missing observations within the modelling window. U.S. commercial banks report under a common regulatory and accounting framework (Basel III as implemented by the Federal Reserve, and U.S. generally accepted accounting principles, GAAP), which supports pooling across institutions.

The panel is constructed in four sequential phases (Table 1). In Phase A, the raw extraction (799 rows $\times$ 31 columns) is cleaned: the Liquidity Coverage Ratio (82.0% missing) and Net Stable Funding Ratio (97.7% missing) are removed; missing values are reconstructed from accounting identities where an exact formula exists (for example, Net Interest Margin is reconstructed as Net Interest Income divided by Earning Assets, recovering 35 bank-quarters with a correlation of 0.9475 against reported values); and residual gaps caused by delayed regulatory reporting are filled forward within each bank using only that bank's prior-quarter value. ROE, ROA, Net Income, Total Assets and Common Equity are excluded from forward-filling. In Phase B, the bank panel is merged with the FRED series by calendar quarter and the window is fixed at 2016Q4–2026Q1 (798 rows $\times$ 37 columns, 21 banks, no duplicate bank-quarters). Phase C engineers 16 features (Section 3.3). In Phase D, the target is constructed by shifting each bank's ROE forward one quarter within its own series, which creates one structural gap per bank and reduces the usable sample to 777 observations.

Raymond James Financial Corp. is excluded from the forecasting comparison because its credit-risk variables are structurally unavailable for a large share of the sample. Removing its 37 observations from the 777-observation file leaves 740; complete-case filtering removes a further 60 bank-quarter observations, yielding a Stage 1 sample of 680 observations for 20 banks, with the estimation sample spanning 2017Q1–2025Q4. The sample covers materially different regimes: expansion (2017–2019), the COVID-19 shock (2020), recovery (2021–2022) and the most rapid U.S. monetary tightening in four decades, followed by the transition to easing.

Table 1. Sample construction and data coverage.

Step	Procedure	Observations
Raw extraction	LSEG bank-level data, 21 banks, 29 financial-statement variables	799 rows $\times$ 31 cols
Phase A	Removal of LCR and NSFR; identity-based reconstruction; within-bank forward-fill	799 rows
Phase B	Merge with 8 FRED series; window fixed at 2016Q4–2026Q1 (one boundary row removed)	798 rows $\times$ 37 cols
Phase C	16 engineered features (lags, growth rates, interactions)	798 rows
Phase D	One-quarter-ahead ROE target; 21 structural end-of-sample gaps removed	777
Exclusion	Raymond James Financial Corp. (37 observations)	740
Stage 1 sample	Complete-case filtering (60 observations removed)	680 (20 banks)
Training	2017Q1–2023Q4	520
Test	2024Q1–2025Q4 (8 quarters)	160
Stage 2 training	Valuation-specific LightGBM; 9 banks; 70 engineered variables $\rightarrow$ 20 features	342
Stage 2 valuation	JPMorgan Chase & Co., rolling valuation dates 2024Q1–2026Q1	9 dates

Notes: Bank data from LSEG Workspace; macroeconomic data from FRED.

For the valuation stage, JPMorgan Chase & Co. is selected because it provides complete firm-specific book value, dividend, market price and forecasting data across all rolling valuation dates. The nine dates

(2024Q1–2026Q1) span the end of monetary tightening and the start of easing, a period in which future bank profitability was materially uncertain.

3.2. Research design

The empirical analysis proceeds in two stages that answer separate questions. *Stage 1* compares econometric and machine-learning models for one-quarter-ahead ROE forecasting under a common chronological train–test design (RQ1). *Stage 2* asks whether a forecasting architecture informed by Stage 1 improves valuation accuracy when incorporated into an RIV framework (RQ2). The Stage 2 architecture is not a reuse of the Stage 1 model: it is a valuation-specific LightGBM system that generates a 20-quarter profitability path for a single bank. Accordingly, Stage 2 evaluates whether the Enhanced forecasting architecture improves RIV performance, rather than isolating the contribution of the LightGBM algorithm itself.

3.3. ROE forecasting framework and variable construction

The forecasting target is one-quarter-ahead ROE, $ROE_{i,t+1}$. Phase C engineers 16 features within each bank’s chronologically sorted series: ten one- and two-quarter lags (of which ROE_Lag1 is the strongest single predictor), four quarter-on-quarter growth rates (assets, deposits, loans and equity), and two interaction terms. The interaction of Net Interest Margin with the Federal Funds rate is motivated by evidence that rising rates disproportionately benefit banks with already elevated margins (Borio et al., 2017); the interaction of loan loss provisions with GDP growth is motivated by evidence that credit stress is amplified when it coincides with weak economic conditions (Climent et al., 2019).

For Stage 1, the resulting pool of 51 candidate predictors (27 bank-level, 8 macroeconomic and 16 engineered variables) is screened using missing-value review, Pearson correlation diagnostics (140 pairs with $|r| \geq 0.80$, mainly among dollar-scale size and income variables), variance inflation factors (VIF; 42 of 51 variables with $VIF > 10$), predictive-relevance measures (Pearson and Spearman correlation and mutual information with the target), scale comparability and economic reasoning. Diagnostics inform, rather than mechanically determine, exclusion: collinear accounting identities and near-duplicate balance-sheet measures are removed, and dollar-denominated variables are excluded where ratio equivalents exist.

Two purpose-specific specifications result (Table 2). *Stage 1* reduces the 51 candidates to a common set of 14 numerical predictors, so that differences in accuracy can be attributed to modelling approach rather than to information sets; the machine-learning models additionally receive 20 bank-identity indicators (34 processed input columns). *Stage 2* uses a separate valuation-specific pipeline: a broader set of 70 engineered variables—which adds dynamically generated ROE and ROE-spread history features to bank fundamentals and macroeconomic variables—is reduced to 20 features covering ROE history, ROE-spread history, bank fundamentals and macroeconomic conditions. Rolling ROE statistics (4- and 8-quarter moving averages and 4-quarter volatility) and ROE-spread history features are constructed only within the relevant forecasting procedures, using information available at each estimation or valuation date. The two specifications serve different empirical purposes and are not successive stages of a single reduction pathway.

Table 2. Predictor definitions and specifications.

Group	Variable	Description / rationale	Stage 1 corr. r	Stage 2 exp. sign
ROE history	ROE_Lag1	One-quarter lag of ROE; profitability persistence	+0.621	+
	ROE_Lag2	Two-quarter lag; medium-term momentum	N/A	+
	ROE_Lag4	Year-ago ROE; seasonal signal	N/A	+
	ROE_MA_4Q, ROE_MA_8Q	4- and 8-quarter moving averages of ROE	N/A	+
	ROE_STD_4Q	4-quarter ROE volatility	N/A	–
Spread history	ROE_SPREAD_Lag1, _Lag2	Lagged ROE less cost of equity (abnormal profitability)	N/A	+
	ROE_SPREAD_MA_4Q	4-quarter moving average of the ROE spread	N/A	+
Interest margin	Net Interest Margin (%)	Core margin; monetary-policy channel	+0.099	+
Credit risk	LLP	Provision & impairment for loan losses	–0.033	N/A
	Net Charge-Off Rate (%)	Realised credit losses	–0.105	–
	LLP $\times$ GDP	Provisions interacted with GDP growth	+0.065	N/A
	LLP_to_Assets	Scale-normalised provisioning rate	N/A	–
Size	Log_Total_Assets	Log of total assets	n.r.	N/A
Capital	Core Tier 1 Ratio (%)	Regulatory capital strength	+0.254	+
Efficiency	Efficiency Ratio (%)	Operating cost relative to income	–0.466	–
Growth	Loan_Growth	Quarter-on-quarter loan growth	+0.130	+ / –
Macroeconomic	GDP_Growth	Real GDP growth (QoQ, %)	+0.029	+
	Fed_Funds_Rate	Effective Federal Funds rate	+0.100	+
	Yield_Spread_10Y_2Y	10-year less 2-year Treasury yield	–0.139	+
	SLOOS_Lending_Standards	Net tightening of lending standards	+0.044	–
	SLOOS_Loan_Demand	Net loan demand	–0.073	+
Number of predictors			14	20

Notes: Stage 1 corr. r is the Pearson correlation of each Stage 1 predictor with one-quarter-ahead ROE. Stage 2 exp. sign is the expected direction of the relationship with future ROE in the 20-feature valuation specification: + positive, – negative, +/– horizon-dependent (positive near-term volume effect; credit-risk indicator over longer horizons). N/A: variable not included in that specification. n.r.: included in Stage 1, correlation not reported.

3.4. Forecasting models

Twelve specifications are estimated under a common target and test period, with preprocessing adapted where the modelling approach requires it.

Econometric panel models. Three specifications are estimated for bank i in quarter t , where $X_{i,t}$ is the vector of 14 predictors:

$$\text{Pooled OLS: } ROE_{i,t+1} = \alpha + \beta' X_{i,t} + \varepsilon_{i,t}, \quad (1)$$

$$\text{Fixed Effects: } ROE_{i,t+1} = \alpha_i + \beta' X_{i,t} + \varepsilon_{i,t}, \quad (2)$$

$$\text{Random Effects: } ROE_{i,t+1} = \alpha + \beta' X_{i,t} + u_i + \varepsilon_{i,t}. \quad (3)$$

Pooled OLS (POLS) ignores bank identity; Fixed Effects removes each bank's mean so that coefficients reflect within-bank variation; Random Effects is more efficient than Fixed Effects if bank effects are uncorrelated with the regressors. The Hausman (1978) test adjudicates between Fixed and Random Effects for coefficient interpretation, and Driscoll–Kraay (1998) standard errors correct for heteroskedasticity, serial correlation and cross-sectional dependence.

Machine-learning models. Nine models of increasing flexibility are estimated: unregularised Linear Regression; Ridge, Lasso and Elastic Net (Zou & Hastie, 2005) regressions; a single Decision Tree; Random Forest (Breiman, 2001), which averages bootstrapped trees; and three gradient-boosting implementations—XGBoost (Chen & Guestrin, 2016), CatBoost, which handles categorical features such as bank identity

natively (Prokhorenkova, Gusev, Vorobev, Dorogush, & Gulin, 2018), and LightGBM (Ke et al., 2017). Numerical predictors are standardised using the training-period mean and standard deviation only. Hyperparameters are selected by five-fold expanding-window (time-aware) cross-validation on the training data only (Table 6).

LightGBM. LightGBM is a gradient-boosting method: trees are added sequentially, each fitted to the residual errors of the current ensemble, so that the model accumulates nonlinear relationships and interactions among predictors without prespecified functional forms. It uses histogram-based split finding and leaf-wise tree growth. The selected configuration uses 63 leaves, a learning rate of 0.03 and a feature fraction of 0.7. Overfitting is controlled through a minimum leaf-sample requirement, L2 regularisation, and feature and row subsampling. Models are trained once on the chronological training window and evaluated out of sample on the subsequent test window.

Stage 2 forecasting architecture. The Enhanced RIV uses a valuation-specific LightGBM system trained on 342 bank-quarter observations from nine banks, using the 20 features selected from 70 engineered variables. It generates a 20-quarter profitability path in three phases. For quarters 1–4, separate horizon-specific LightGBM models produce direct forecasts, preventing errors at earlier horizons from propagating mechanically into later near-term forecasts. For quarters 5–8, a continuation model is applied recursively using the previously estimated ROE spread, a rolling average of recent predicted spreads, and dynamically updated bank-specific and macroeconomic inputs. From quarter 9 onward, abnormal profitability converges through geometric persistence decay,

$$s_{t_0+h} = \tilde{\omega} s_{t_0+h-1}, \quad \tilde{\omega} = \min(\hat{\omega}, 0.92), \quad h = 9, \dots, 20, \quad (4)$$

where $s = ROE - r_q$ is the quarterly ROE spread, $\hat{\omega}$ is the estimated persistence parameter and 0.92 is the maximum persistence parameter, imposed to prevent an unusually high estimate from implying an implausibly long duration of abnormal profitability. The hybrid structure reflects increasing uncertainty at longer horizons: near-term forecasts rely most directly on observed information, whereas medium-term projections increasingly depend on estimated inputs.

3.5. Residual income valuation framework

Under clean surplus accounting, book value evolves as

$$BV_t = BV_{t-1} + NI_t - D_t, \quad (5)$$

where NI_t is net income and D_t dividends, and intrinsic value equals current book value plus the present value of expected residual income (Ohlson, 1995),

$$V_0 = BV_0 + \sum_{t=1}^{\infty} \frac{E_0[RI_t]}{(1+r_e)^t}. \quad (6)$$

Implementation. Both valuation models are implemented quarterly over an explicit horizon of 20 quarters from valuation date t_0 . Quarterly residual income is forecast net income less the equity charge on opening book value,

$$RI_{t_0+h} = NI_{t_0+h} - r_q BV_{t_0+h-1} = (ROE_{t_0+h} - r_q) BV_{t_0+h-1}, \quad (7)$$

where r_q is the quarterly cost of equity and ROE_{t_0+h} is forecast quarterly profitability. Book value per share

is projected under Equation (5) from forecast net income and a payout ratio estimated from JPMorgan's trailing eight-quarter history, so that clean surplus holds at every step. Intrinsic value per share is

$$V_{t_0} = BV_{t_0} + \sum_{h=1}^{20} \frac{RI_{t_0+h}}{(1+r_q)^h} + \frac{TV_{20}}{(1+r_q)^{20}}, \quad (8)$$

where the terminal value capitalises quarter-20 residual income as a perpetuity with growth g ,

$$TV_{20} = \frac{RI_{t_0+20}(1+g)}{r_q - g}, \quad g = 0 \Rightarrow TV_{20} = \frac{RI_{t_0+20}}{r_q}. \quad (9)$$

The terminal value is therefore a no-growth perpetuity. The persistence parameter does not enter Equation (9); it affects terminal value only indirectly, through the ROE-spread path that determines RI_{t_0+20} .

Traditional RIV. The benchmark projects ROE with a fixed persistence rate toward the cost of equity (CoE) over the full horizon,

$$ROE_{t+1} = CoE + \omega (ROE_t - CoE), \quad (10)$$

where ω is estimated at each valuation date by regressing the abnormal ROE spread on its one-quarter lag using information then available, constrained to $[0, 1]$.

Enhanced RIV. The Enhanced model retains the accounting structure, cost-of-equity estimation, clean-surplus projection, discounting and terminal-value formula of the Traditional model and replaces only the mechanism generating the ROE path with the three-phase architecture of Section 3.4: direct LightGBM forecasts for quarters 1–4, a recursive continuation for quarters 5–8 and capped geometric decay (Equation 4) for quarters 9–20. Both models thus convert their profitability paths into value through identical formulae; they differ in the quarter-20 residual income that enters Equation (9), because that quantity depends on each model's forecast path, including the capped persistence parameter applied in the Enhanced model over quarters 9–20. The Enhanced model does not forecast the share price; it forecasts the profitability path from which residual income and intrinsic value are derived.

3.6. Cost of equity estimation

The cost of equity is estimated at each valuation date with the CAPM,

$$r_e = r_f + \beta \times ERP, \quad (11)$$

where r_f is the prevailing three-month U.S. Treasury bill yield, β is an expanding-window equity beta estimated against the S&P 500, and ERP is the annual implied equity risk premium from Damodaran's series (Damodaran, 2024). The resulting rate is held constant across the 20-quarter horizon for each valuation date, and its quarterly equivalent r_q is used for the equity charge and for discounting. Because the same CAPM estimate enters both models, any bias arising from CAPM misspecification affects both equally.

3.7. Forecast evaluation and statistical testing

The Stage 1 panel is split chronologically rather than randomly: models are trained on 2017Q1–2023Q4 (520 observations) and tested on 2024Q1–2025Q4 (160 observations). Parameters are estimated once and held fixed throughout the test window; only the realised lagged inputs update each quarter. Accuracy is measured by root mean squared error (RMSE) and the coefficient of determination (R^2) on the test set, and overfitting by the generalisation gap, defined as test RMSE minus training RMSE.

Differences in predictive accuracy are assessed with the Diebold and Mariano (1995; DM) test. For forecast errors $e_{1,t}$ and $e_{2,t}$ of two competing models and loss function $L(\cdot)$, the loss differential is $d_t = L(e_{1,t}) - L(e_{2,t})$ and

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}}, \quad (12)$$

which is asymptotically standard normal under the null of equal predictive accuracy. Sixteen pairwise tests compare each of the four ensemble models with the Decision Tree, Elastic Net, Pooled OLS and Fixed Effects benchmarks; a further test compares Pooled OLS with Fixed Effects. The analysis distinguishes throughout between point-estimate performance (a lower test RMSE) and statistically significant predictive superiority (rejection of the Diebold–Mariano null).

3.8. Valuation accuracy evaluation

Stage 2 uses an expanding-window design: at each valuation date the training sample includes only observations whose target quarter falls strictly before that date, and the cost of equity, persistence coefficient and book value are estimated from information available up to the preceding quarter. Intrinsic values are compared with observed market prices using

$$MAE = \frac{1}{N} \sum_{j=1}^N |V_j - P_j|, \quad RMSE = \sqrt{\frac{1}{N} \sum_{j=1}^N (V_j - P_j)^2}, \quad MAPE = \frac{100}{N} \sum_{j=1}^N \left| \frac{V_j - P_j}{P_j} \right|, \quad (13)$$

with $N = 9$ valuation dates. Market price is used as a transparent empirical benchmark, not as an error-free measure of unobservable intrinsic value.

4. Empirical results

4.1. Descriptive statistics

Table 3 reports descriptive statistics for the target and key predictors. One-quarter-ahead ROE has a mean of 11.45% and a median of 11.02%, with pronounced positive skewness (6.84) and excess kurtosis (77.97) (Figure 1). The upper tail reflects First Citizens BancShares' 2023 acquisition-related accounting gain, which raised quarterly ROE to approximately 78%; the lower tail includes Truist Financial Corporation's 2024 goodwill impairment (−12.67%). Twenty-eight observations (3.6% of the sample) lie outside the conventional $1.5 \times$ interquartile-range (IQR) fences. They are retained without winsorisation because each corresponds to a reported outcome linked to an identifiable economic event—pandemic-era provisioning, goodwill impairments, securities losses, rate-cycle effects and acquisition-related gains. Among the predictors, the Efficiency Ratio is highly dispersed, and GDP growth spans an unusually wide range of macroeconomic conditions.

Profitability differs persistently across banks (Table A1). Raymond James (median ROE 17.02%) and East West Bancorp (15.66%) rank consistently at the upper end of the distribution, whereas Citizens Financial (7.43%) and Truist (8.42%) rank at the lower end, a gap of approximately 10 percentage points. This heterogeneity is consistent with the case for bank-specific effects. Over time, cross-bank average profitability rises gradually before the pandemic, contracts sharply in 2020, recovers, peaks during the monetary tightening period and moderates toward the end of the sample (Figure 2). These regime-dependent movements motivate the inclusion of macroeconomic predictors and the evaluation of flexible forecasting models.

Table 3. Descriptive statistics for the target and key predictors ($n = 777$).

Variable	Mean	Median	Std. dev.	Min	Max	Skewness
ROE_{t+1} (% , target)	11.45	11.02	5.83	-12.67	78.36	6.84
Net Interest Margin (%)	3.03	3.08	0.51	0.42	4.22	-0.59
Core Tier 1 Ratio (%)	11.41	10.77	2.63	4.50	25.90	2.99
Efficiency Ratio (%)	57.09	58.16	32.02	-761.68	197.44	-21.88
Net Charge-Off Rate (%)	0.28	0.24	0.21	-0.20	1.51	1.28
GDP growth (QoQ, %)	2.54	2.70	5.91	-28.00	34.90	-0.42
Federal Funds rate (%)	2.32	1.94	1.93	0.06	5.33	0.38

Notes: Statistics are computed on the 777-observation file for 21 banks, which precedes the exclusion of Raymond James Financial Corp. and complete-case filtering; the Stage 1 estimation sample contains 680 observations. Excess kurtosis of the target is 77.97. QoQ: quarter-on-quarter. Per-bank statistics are reported in Table A1.

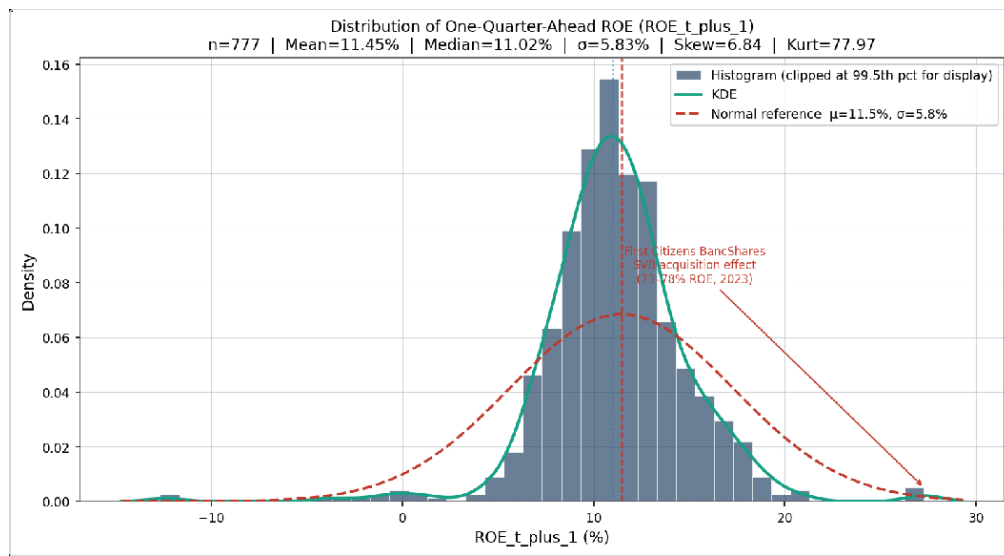


Figure 1. Distribution of one-quarter-ahead bank ROE ($n = 777$). The histogram is clipped at the 99.5th percentile for display; the kernel density estimate and a normal reference distribution with the same mean and standard deviation are overlaid.

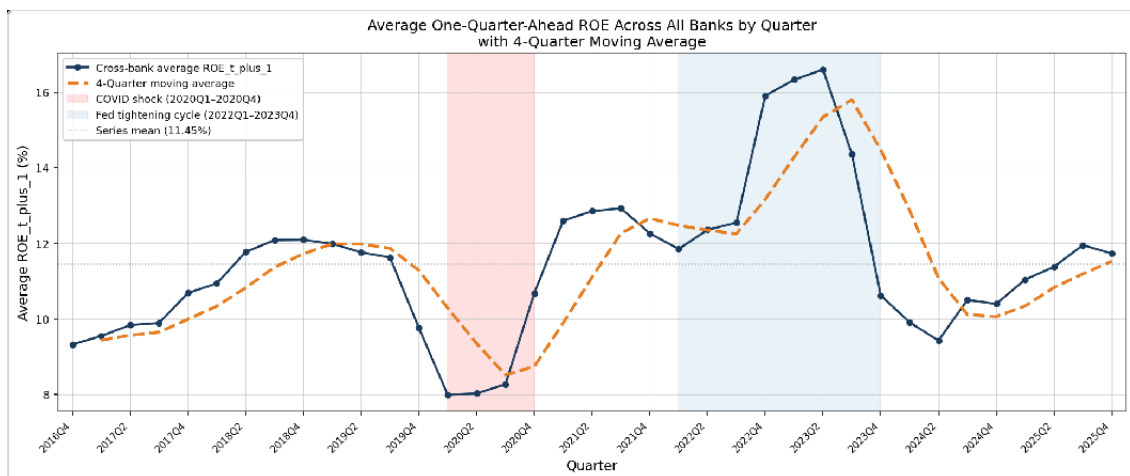


Figure 2. Cross-bank average one-quarter-ahead ROE, 2016Q4–2025Q4, with four-quarter moving average. Shaded areas mark the COVID-19 shock and the Federal Reserve tightening cycle.

4.2. Econometric forecasting results

Table 4 reports coefficient estimates for the three panel specifications on the full 680-observation sample. Random Effects coincides numerically with Pooled OLS at the reported precision because the estimated between-bank variance component is small relative to the within-bank variance; the specifications are nonetheless reported separately because their standard-error structures and consistency properties differ. Breusch–Pagan (Breusch & Pagan, 1979) and White (1980) tests reject homoskedasticity, a Wooldridge test detects serial correlation (Wooldridge, 2010), and a Pesaran CD test detects cross-sectional dependence (Pesaran, 2021), which motivates Driscoll–Kraay standard errors for inference on the Fixed Effects specification.

The Hausman test rejects the null that bank effects are uncorrelated with the regressors ($\chi^2(14) = 44.208$, $p = 0.000055$), so Fixed Effects is the preferred estimator for coefficient interpretation. Under Fixed Effects, seven of the fourteen predictors are significant at the 5% level or better (Table 4): ROE_Lag1, Net Interest Margin, loan loss provisions, the Net Charge-Off Rate, Log_Total_Assets, the Efficiency Ratio and Loan_Growth. Dropping the most collinear predictor and comparing Fixed Effects coefficient signs with an independent Ridge regression do not alter this conclusion. Under Fixed Effects, the coefficient on ROE_Lag1 (0.464) indicates substantial within-bank persistence: approximately 46% of a one-point deviation in current ROE carries into the next quarter. The Efficiency Ratio (−0.075) and Net Charge-Off Rate (−2.639) enter with the expected negative signs. None of the five macroeconomic predictors is individually significant, despite substantial correlation among them (the Federal Funds rate and the 10Y–2Y spread correlate at −0.76). This pattern is consistent with macroeconomic variables operating jointly through the interest-rate cycle, which a linear specification with individually estimated coefficients may not capture.

Table 4. Panel regression estimates for one-quarter-ahead ROE.

Variable	Pooled OLS		Fixed Effects		Random Effects	
	Coef.	SE	Coef.	SE	Coef.	SE
ROE_Lag1	0.5549***	0.0504	0.4638***	0.0387	0.5549***	0.0504
Net Interest Margin (%)	0.6303	0.4950	2.0050***	0.6982	0.6303	0.4950
Provision & impairment (LLP)	−0.0006**	0.0003	−0.0008**	0.0003	−0.0006**	0.0003
Net Charge-Off Rate (%)	−2.2600**	0.9512	−2.6394**	0.8243	−2.2600**	0.9512
Log_Total_Assets	0.4016***	0.1483	1.8977***	0.7782	0.4016***	0.1483
Core Tier 1 Ratio (%)	0.3224***	0.1087	−0.1690	0.2825	0.3224***	0.1087
Efficiency Ratio (%)	−0.0780***	0.0014	−0.0753***	0.0013	−0.0780***	0.0014
Loan_Growth	−0.0340*	0.0087	−0.0401**	0.0156	−0.0340*	0.0087
LLP × GDP	0.0000	0.0000	−0.0000	0.0000	0.0000	0.0000
GDP_Growth	0.0143	0.0102	0.0218	0.0079	0.0143	0.0102
Fed_Funds_Rate	−0.2112	0.2196	−0.2269	0.2521	−0.2112	0.2196
Yield_Spread_10Y_2Y	−0.0378	0.7178	0.0310	0.7063	−0.0378	0.7178
SLOOS_Lending_Standards	0.0115	0.0099	0.0019	0.0066	0.0115	0.0099
SLOOS_Loan_Demand	0.0004	0.0055	−0.0076	0.0089	0.0004	0.0055
R^2 (within R^2 for FE)	0.5669		0.4858		0.5669	
Observations	680		680		680	

Hausman test (FE vs. RE): $\chi^2(14) = 44.208$; $p = 0.000055$; Fixed Effects preferred for coefficient interpretation.

Notes: Sample of 20 U.S. commercial banks, 2017Q1–2025Q4. FE: Fixed Effects; RE: Random Effects; LLP: provision & impairment for loan losses. SE denotes robust standard errors; inference for the Fixed Effects specification uses Driscoll–Kraay standard errors. ***, ** and * denote significance at the 1%, 5% and 10% levels.

Forecasting accuracy is a separate question from estimator consistency. On the chronological hold-

out period, Pooled OLS (equivalently Random Effects) records a test RMSE of 4.399 and Fixed Effects 4.889 (Table 5). Pooled OLS produces significantly lower forecast errors than Fixed Effects (Diebold–Mariano statistic = -2.340 , $p = 0.021$). This does not contradict the Hausman result: the Hausman test concerns consistency for coefficient interpretation, whereas the Diebold–Mariano test concerns relative out-of-sample accuracy. One possible explanation is that bank-specific intercepts estimated on 2017–2023 data adapted poorly to the subsequent regime transition. Bank-level RMSE under Fixed Effects ranges from 0.89 (Regions Financial) to 17.83 (First Citizens BancShares). The two worst-forecast banks illustrate opposite failure modes of a lag-driven linear model: First Citizens is over-predicted because its inflated 2023 value is extrapolated forward, whereas Truist is under-predicted for several quarters after its 2024 impairment because ROE_Lag1 anchors each forecast to the preceding depressed quarter. By quarter, the largest error occurs in 2024Q1 (RMSE = 10.3), when lagged inputs still reflect the preceding regime. Pooled OLS therefore serves as the econometric forecasting benchmark.

Table 5. Out-of-sample forecast performance of the econometric panel models.

Model	Test RMSE	Test R^2	Generalisation gap
Pooled OLS / Random Effects	4.399	-0.053	$+0.440$
Fixed Effects	4.889	-0.300	$+1.155$

Notes: Training period 2017Q1–2023Q4 ($n = 520$); test period 2024Q1–2025Q4 ($n = 160$). Generalisation gap = test RMSE – training RMSE. Pooled OLS and Random Effects produce numerically identical forecasts. Diebold–Mariano test, Pooled OLS vs. Fixed Effects: $DM = -2.340$, $p = 0.021$.

4.3. Machine-learning forecasting results

The machine-learning models use the same 14 numerical predictors, target and chronological split, with bank identity represented by 20 one-hot indicators. The test period represents a different macroeconomic environment from the training period. Several macroeconomic predictors, most clearly the Federal Funds rate, shift in distribution between the two windows (Figure A1), and a two-sample Kolmogorov–Smirnov test rejects equality of the training and test ROE distributions ($p = 0.034$); mean ROE declines from 11.71% to 10.42%. The hold-out evaluation therefore takes place under temporal distribution shift.

Table 6 reports the tuned hyperparameters of the ensemble models, and Table 7 reports in-sample and out-of-sample performance. Among the baseline models, unregularised Linear Regression records the lowest training RMSE (3.734) but the highest test RMSE (4.889), indicating poor generalisation with 34 inputs and 520 training observations; its test RMSE is effectively identical to that of Fixed Effects (4.889 versus 4.8885), consistent with both combining bank-specific effects with a common linear response. Ridge, Lasso and Elastic Net accept modestly higher training error in exchange for lower test error (generalisation gaps of 0.195–0.387). The Decision Tree records the lowest test RMSE among the baseline models (4.036). Four of the five baseline models outperform the Pooled OLS benchmark.

Table 6. Hyperparameter selection for the ensemble models.

Model	Search	Selected parameters	CV RMSE
Random Forest	Grid search	max_depth = None; min_leaf = 4; n_estimators = 200	4.968
XGBoost	Randomised search (40)	max_depth = 7; learning rate = 0.01; n_estimators = 200	4.734
LightGBM	Randomised search (40)	num_leaves = 63; learning rate = 0.03; feature_fraction = 0.7	4.778
CatBoost	Randomised search (40)	depth = 7; learning rate = 0.05; iterations = 100	4.667

Notes: CV RMSE: cross-validation RMSE. Hyperparameters are selected by time-aware (expanding-window) cross-validation on the training period only. The number in parentheses is the number of randomised-search candidates.

Table 7. In-sample and out-of-sample performance of the machine-learning models.

Model	Train RMSE	Test RMSE	Test R^2	Gen. gap
<i>Panel A: Baseline models</i>				
Linear Regression	3.734	4.889	−0.300	+1.155
Ridge Regression	3.843	4.230	0.027	+0.387
Lasso Regression	3.961	4.191	0.045	+0.230
Elastic Net	3.960	4.156	0.060	+0.195
Decision Tree	4.852	4.036	0.114	−0.816
<i>Panel B: Ensemble models</i>				
Random Forest	3.912	3.461	0.349	−0.452
XGBoost	3.632	3.648	0.276	+0.016
LightGBM	4.531	3.308	0.405	−1.223
CatBoost	1.433	3.500	0.334	+2.066

Notes: 34 inputs (14 standardised numerical predictors and 20 bank indicators). Training $n = 520$; test $n = 160$. Gen. gap = test RMSE – training RMSE.

All four ensemble models outperform the Pooled OLS benchmark. LightGBM records the lowest test RMSE (3.308) and the highest test R^2 (0.405), followed by Random Forest, CatBoost and XGBoost. Cross-validation rankings did not transfer to the hold-out period: CatBoost achieved the lowest cross-validation RMSE (4.667) but the largest positive generalisation gap (+2.066), whereas LightGBM ranked third in cross-validation. This contrast underlines the importance of evaluation on a genuinely unseen chronological window.

LightGBM's test error is lower than its training error, producing a negative generalisation gap (−1.223). This should not be read as evidence that the model performs intrinsically better on unseen data. A more cautious interpretation is that the training and test periods differ in difficulty and distribution, as the Kolmogorov–Smirnov result indicates, and that the low learning rate and subsampling in the selected configuration limit how closely individual trees fit the training data. Evidence from a single hold-out period cannot determine whether the negative gap would persist under alternative test windows.

No ensemble model eliminates the influence of extreme ROE observations. Test residuals remain positively skewed and leptokurtic for all four ensembles (Table 8), with the largest errors concentrated around the tail observations identified in Section 4.1. LightGBM has the smallest residual standard deviation (3.191) and a comparatively concentrated residual distribution over the ordinary ROE range (Figure 3). The remaining non-normality is interpreted as unresolved difficulty in forecasting tail events; because the objective is out-of-sample prediction, it is not a reason to discard the model.

Table 8. Distribution of test-period residuals for the ensemble models.

Model	Mean	Std. dev.	Skewness	Kurtosis
Random Forest	1.046	3.299	3.96	19.9
XGBoost	1.077	3.486	3.06	11.12
LightGBM	0.874	3.191	3.136	13.12
CatBoost	0.818	3.403	2.932	10.5

Notes: Residuals are computed on the 160 test observations (2024Q1–2025Q4).

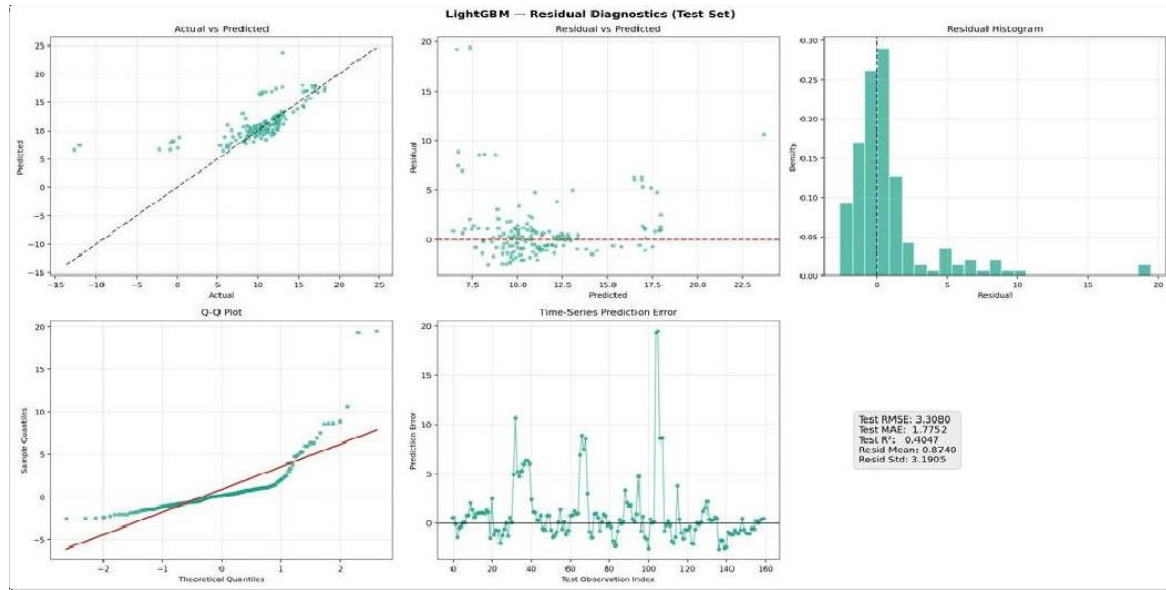


Figure 3. Residual diagnostics for LightGBM on the test set (2024Q1–2025Q4): actual versus predicted ROE, residuals versus predicted values, residual histogram, normal Q–Q plot and prediction error by test observation.

4.4. Model comparison

Twelve specifications were estimated. Because Random Effects and Pooled OLS produce identical forecasts, they yield eleven distinct model forecasts. Table 9 groups models with identical reported test metrics into a single ranking row, so the eleven distinct forecasts occupy ten rows: Pooled OLS and Random Effects share a row because their forecasts are identical, and Fixed Effects and unregularised Linear Regression share a row because their reported test RMSE (4.889) and test R^2 (−0.300) are identical, although their forecasts are distinct. Eight of the nine machine-learning models outperform Pooled OLS, while unregularised Linear Regression ties with Fixed Effects. The improvement is graduated: approximately 4–6% for the regularised linear models, 8% for the Decision Tree and 17–25% for the ensemble models. This progression is consistent with forecast gains arising incrementally from regularisation, nonlinear partitioning and, for the ensembles, variance reduction or sequential error correction.

Table 9. Overall out-of-sample ranking of forecasting models.

Rank	Model	Category	Test RMSE	Test R^2	Δ RMSE vs. POLS
1	LightGBM	Ensemble	3.308	0.405	−24.8%
2	Random Forest	Ensemble	3.461	0.349	−21.3%
3	CatBoost	Ensemble	3.500	0.334	−20.4%
4	XGBoost	Ensemble	3.648	0.276	−17.1%
5	Decision Tree	Baseline ML	4.036	0.114	−8.3%
6	Elastic Net	Baseline ML	4.156	0.060	−5.5%
7	Lasso Regression	Baseline ML	4.191	0.045	−4.7%
8	Ridge Regression	Baseline ML	4.230	0.027	−3.8%
9	Pooled OLS / Random Effects	Econometric	4.399	−0.053	–
10	Fixed Effects / Linear Regression	Econometric / ML	4.889	−0.300	+11.1%

Notes: Test period 2024Q1–2025Q4 ($n = 160$). Ten ranking rows cover twelve specifications and eleven distinct forecasts; rows 9 and 10 each group two specifications with identical reported test metrics. POLS: Pooled OLS. Δ RMSE is the percentage difference in test RMSE relative to Pooled OLS (4.399), computed from the reported values. Rankings are point estimates; see Table 10 for tests of predictive accuracy.

Forecast errors are largest at the beginning of the test period and narrow thereafter (Figure 4). This does not reflect model adaptation: all models are estimated once on data through 2023Q4 and held fixed.

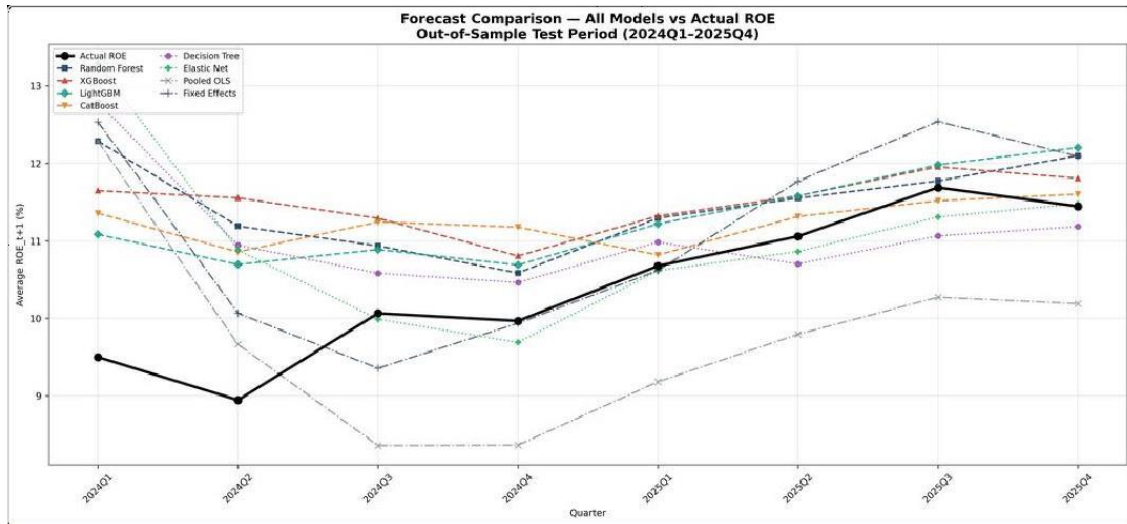


Figure 4. Average actual and predicted ROE across banks for all models during the out-of-sample test period, 2024Q1–2025Q4.

From 2024Q2 onward, ROE_Lag1 incorporates realised test-period outcomes rather than the last training-period observation, so forecasts benefit from progressively more current profitability information without re-estimation.

4.5. Diebold–Mariano tests

Table 10 summarises the pairwise tests. Of the sixteen comparisons between ensemble models and benchmarks, only Random Forest versus the Decision Tree rejects the null of equal predictive accuracy at the 5% level. None of LightGBM’s four comparisons is significant, including its comparison with Pooled OLS ($p = 0.487$). Although LightGBM achieves the lowest test RMSE, its superiority over the benchmark models cannot be established statistically from this test period.

The absence of significance does not invalidate the RMSE ranking; it indicates that the evidence is insufficient to distinguish the measured differences from zero. Statistical power is constrained by the structure of the test sample: the 160 observations cover 20 banks over eight quarters and are unlikely to be independent in either dimension, because banks share macroeconomic shocks within a quarter and forecast errors for a given bank may be correlated over time. This is a plausible explanation for the lack of significance rather than an established one.

Table 10. Diebold–Mariano tests of equal predictive accuracy.

Ensemble model	Benchmark model			
	Decision Tree	Elastic Net	Pooled OLS	Fixed Effects
LightGBM	n.s.	n.s.	$p = 0.487$	n.s.
Random Forest	$p < 0.05$	n.s.	n.s.	n.s.
XGBoost	n.s.	n.s.	n.s.	n.s.
CatBoost	n.s.	n.s.	n.s.	n.s.

Econometric comparison: Pooled OLS vs. Fixed Effects, $DM = -2.340$, $p = 0.021$ (favours Pooled OLS).

Notes: n.s. denotes failure to reject the null of equal predictive accuracy at the 5% level. Test period 2024Q1–2025Q4 ($n = 160$). Of the sixteen ensemble–benchmark comparisons, only Random Forest versus the Decision Tree rejects the null at the 5% level; the p -value is shown for the LightGBM–Pooled OLS comparison.

4.6. Feature importance and model interpretation

Permutation importance (Breiman, 2001) on the held-out test set (30 repeats) identifies ROE_Lag1 as the dominant predictor in the LightGBM model, followed at a considerable distance by the Efficiency Ratio, the Net Charge-Off Rate, Loan Growth and the Federal Funds rate (Figure 5; Table 11). The dominance of lagged ROE is consistent with the persistence documented in the panel regressions, while the smaller contributions of operating efficiency, credit quality, loan growth and monetary conditions indicate that the model combines the autoregressive signal with additional nonlinear information from bank-specific and macroeconomic variables. These measures describe predictive contributions within the fitted model; they do not identify causal effects of the predictors on bank profitability.

Native (split-based) and permutation importance diverge for some models. Random Forest and XGBoost assign high native importance to the First Citizens BancShares indicator—XGBoost ranks it above ROE_Lag1—but its importance falls substantially under permutation on the test set, indicating that split-based importance overstates the indicator's contribution to out-of-sample accuracy. LightGBM shows greater agreement between native and permutation rankings, which suggests less dependence on a single bank identity.

SHAP (Lundberg & Lee, 2017) is discussed in Section 2.5 as a conceptual framework for interpreting machine-learning forecasts. The empirical interpretation in this study relies on native and permutation importance; SHAP attributions are not estimated or reported, and no SHAP-based results are claimed.

Table 11. LightGBM permutation feature importance on the test set: rank order.

Rank	Feature	Rank	Feature
1	ROE_Lag1	6	Fed_Funds_Rate
2	Efficiency Ratio (%)	7	Provision & impairment for loan losses
3	Net Charge-Off Rate (%)	8	Net Interest Margin (%)
4	Bank indicator: First Citizens BancShares	9	Core Tier 1 Ratio (%)
5	Loan_Growth	10	Yield_Spread_10Y_2Y

Notes: Importance is the increase in test RMSE when a feature is randomly permuted, averaged over 30 repeats. Rank order corresponds to Figure 5, which reports the mean importance and its variability across repeats; the top ten features are listed.

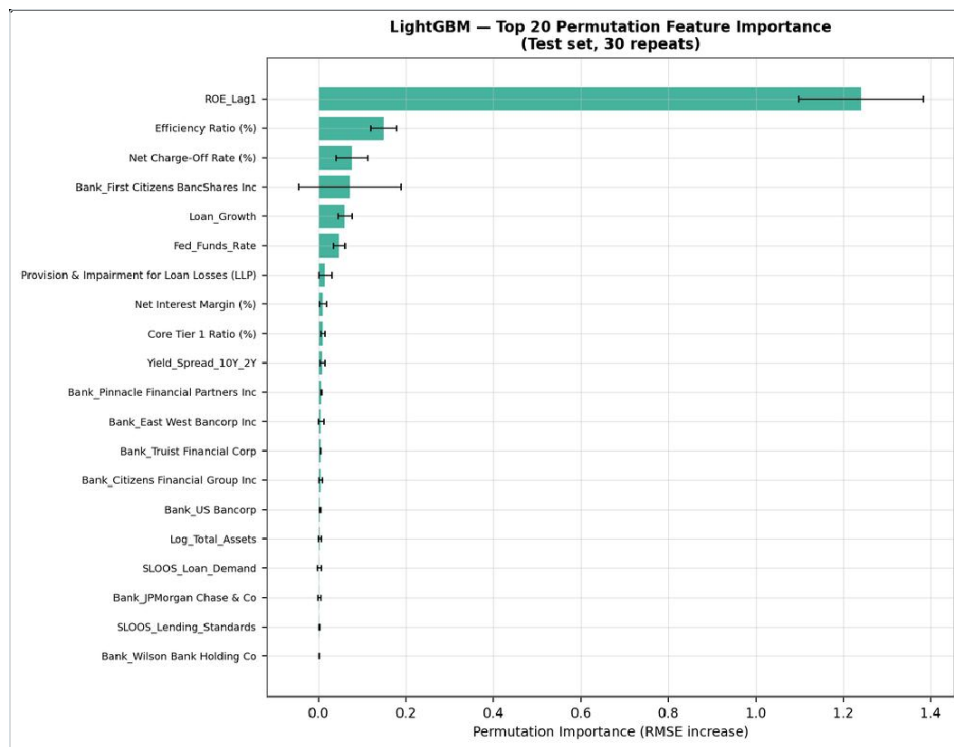


Figure 5. LightGBM permutation feature importance on the held-out test set (top 20 features, 30 repeats). Bars show the mean increase in test RMSE; error bars show variability across repeats.

4.7. Traditional residual income valuation

LightGBM is carried forward to the valuation stage because it records the lowest test RMSE and highest test R^2 , the greatest stability across random-seed re-estimation, and the strongest agreement between native and permutation importance. The choice remains conditional on the observed test window: LightGBM exhibits a negative generalisation gap, and its advantage over the benchmark models is not statistically significant.

The Traditional RIV benchmark is applied to JPMorgan at nine rolling valuation dates. Across these dates, the risk-free rate declines from 5.23% to 3.59% while beta remains between 1.10 and 1.13, reducing the estimated cost of equity from 10.09% to 8.38% (Table 12). The persistence coefficient ω rises from 0.806 to 0.904, although the associated regression R^2 remains modest (0.12–0.20). The benchmark therefore implies persistent abnormal profitability combined with substantial quarter-to-quarter forecast uncertainty.

Table 12. Traditional RIV inputs across valuation dates.

	Risk-free rate	Beta	Cost of equity	Persistence ω	R^2 of ω regression
2024Q1 (first date)	5.23%	1.10–1.13	10.09%	0.806	0.12–0.20
2026Q1 (last date)	3.59%		8.38%	0.904	

Notes: The cost of equity is estimated by CAPM at each valuation date; ω is estimated by regressing the abnormal ROE spread on its one-quarter lag and constrained to $[0, 1]$. The table reports the first and last valuation dates; beta and the R^2 of the persistence regression are reported as ranges across the nine dates.

The 2026Q1 valuation illustrates the book-value anchoring of the framework (Figure 6). Book value contributes \$128.4 per share and the present values of explicit residual income and terminal value contribute \$27.7 and \$21.6, respectively, for an intrinsic value of \$177.7; book value thus accounts for approximately 72% of the estimate. The Traditional estimate lies below the market price at all nine valua-

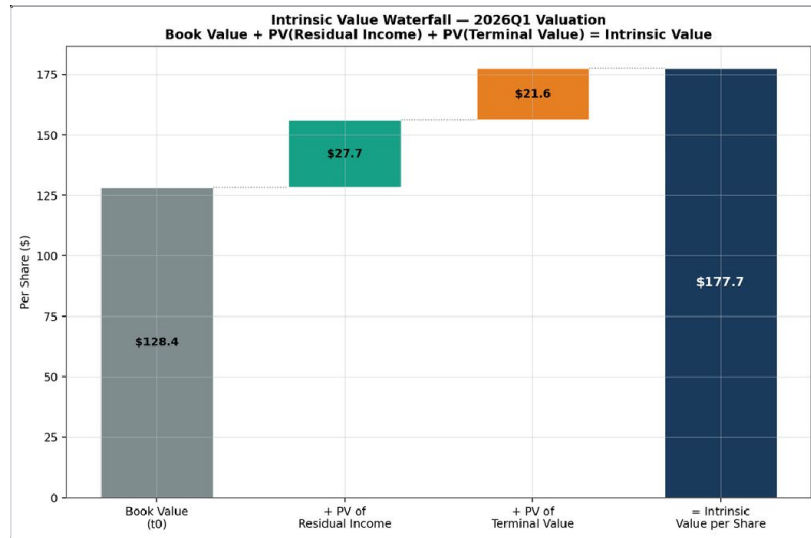


Figure 6. Decomposition of the Traditional RIV intrinsic value for JPMorgan Chase & Co. at 2026Q1 into book value, the present value of explicit-horizon residual income and the present value of terminal value (\$ per share).

tion dates (Table 13), with an MAE of \$115.55 and a MAPE of 44.07%. Although intrinsic value rises over the period, it does not track the magnitude of the market-price increase, and the uniform sign of the errors indicates systematic underestimation rather than symmetric noise. Implementation checks covering temporal alignment, rolling-window construction and clean-surplus reconciliation indicate that the gap is unlikely to arise from these mechanical sources. Plausible contributing mechanisms include the forced convergence of abnormal ROE toward the cost of equity, the inability of a parsimonious benchmark to represent the nonlinear relationships identified in Stage 1, and forward-looking growth expectations in market prices that a book-value-anchored model does not capture.

4.8. Enhanced machine-learning-based RIV

The Enhanced model holds the accounting structure, cost-of-equity estimation, clean-surplus mechanics and terminal-value formula constant and substitutes only the mechanism that generates the ROE path, using the three-phase architecture described in Section 3.4. Each phase addresses a specific problem: direct forecasts are most reliable but cannot be extended over 20 quarters without compounding their own errors as inputs; a purely recursive extension across the full horizon would compound that uncertainty further; and the capped analytical decay imposes economic discipline on the distant horizon, where abnormal profitability cannot plausibly be assumed to persist indefinitely.

4.9. Valuation accuracy comparison

Table 13 and Figure 7 compare the two models with observed market prices, and Table 14 reports the error metrics. The Enhanced model reduces MAE from \$115.55 to \$59.98, RMSE from \$121.25 to \$70.16 and MAPE from 44.07% to 21.91%. Because both models share an identical accounting identity and discounting framework—confirmed by thirteen internal validation checks—the improvement is attributable to the alternative profitability-forecasting architecture and the ROE paths it generates rather than to differences in valuation mechanics. The Enhanced estimate is closer to the market price at almost every date and, at 2024Q3 only, lies slightly above it.

The improvement is not uniform. During 2025, JPMorgan’s share price rose from approximately \$243 to \$328, while the Enhanced intrinsic value remained within a comparatively narrow \$200–\$215 band. The Enhanced model captures a much larger share of the valuation level than the Traditional benchmark,

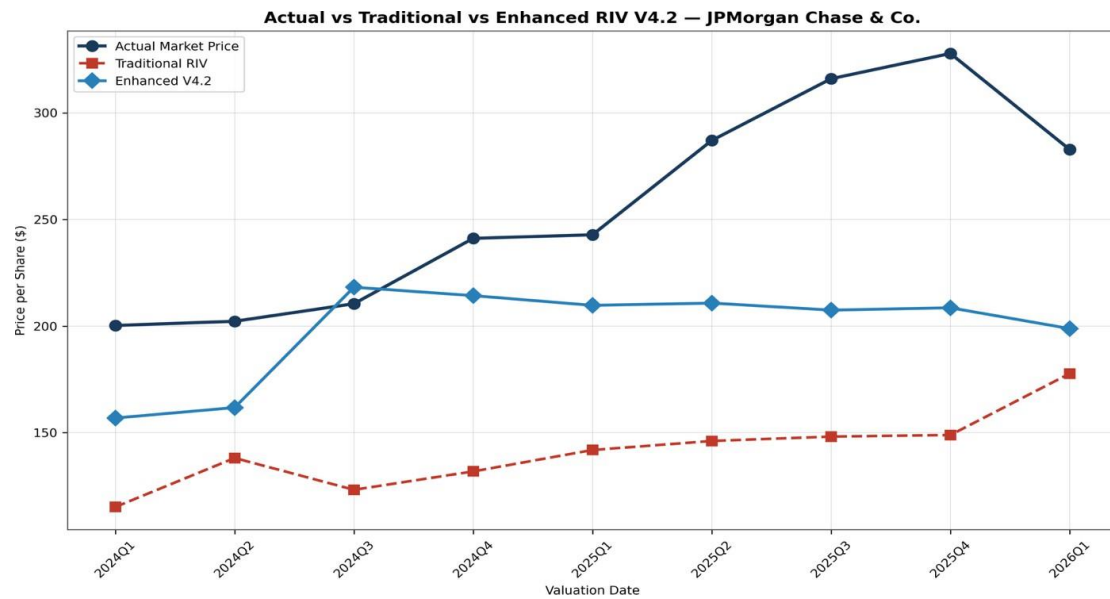


Figure 7. Observed market price and Traditional and Enhanced RIV intrinsic values for JPMorgan Chase & Co. across nine rolling valuation dates, 2024Q1–2026Q1 (\$ per share).

but a substantial gap to market price persists, and at some dates widens, during the period of strongest appreciation. Several explanations are plausible—growth expectations beyond the explicit horizon, franchise value not represented in a book-value-anchored framework, or a market-implied required return different from the CAPM estimate—but none can be confirmed or ruled out with the present evidence.

Table 13. Market price and RIV intrinsic values for JPMorgan Chase & Co. at selected valuation dates (\$ per share).

Valuation date	Market price	Traditional RIV		Enhanced RIV	
		Value	Error (%)	Value	Error (%)
2024Q1	200.30	115.14	−42.5	156.86	−21.7
2024Q3	210.50	123.21	−41.5	218.26	+3.7
2025Q1	242.85	141.87	−41.6	209.73	−13.6
2025Q4	327.91	148.89	−54.6	208.59	−36.4
2026Q1	282.84	177.72	−37.2	198.83	−29.7

Notes: Error = (intrinsic value − market price) / market price. Traditional-model errors are as reported; Enhanced-model errors are computed from the reported values. Five of the nine valuation dates are tabulated; the complete series is shown in Figure 7, and Table 14 reports error metrics over all nine dates.

Table 14. Valuation accuracy of the Traditional and Enhanced RIV models (nine valuation dates).

Model	MAE (\$)	RMSE (\$)	MAPE (%)
Traditional RIV	115.55	121.25	44.07
Enhanced RIV (LightGBM architecture)	59.98	70.16	21.91
Difference	−55.57	−51.09	−22.16 pp
Relative reduction	48.1%	42.1%	50.3%

Notes: Metrics are defined in Equation (13) with $N = 9$ valuation dates (2024Q1–2026Q1). Differences and relative reductions are computed from the reported values. No formal significance test is applied to the valuation comparison.

5. Discussion

5.1. Forecasting evidence

The econometric results show that the preferred model depends on the question asked. Fixed Effects is preferred for coefficient interpretation, yet Pooled OLS forecasts significantly more accurately out of sample; bank-specific intercepts fixed at the end of the training period appear to have anchored forecasts to profitability levels that did not adapt once the monetary-policy environment shifted. Among the machine-learning models, regularisation recovers much of the accuracy an unconstrained linear model loses to overfitting, the Decision Tree improves further by isolating unusual banks in separate branches, and the ensembles extend the pattern, with LightGBM recording the lowest test error. The progression should not be read as evidence that complexity is rewarded automatically: CatBoost fitted its training sample most closely and achieved the best cross-validation score, yet produced the largest positive generalisation gap.

With respect to H1, the evidence supports lower forecast errors for machine-learning models at the level of point estimates: eight of nine machine-learning models improve on Pooled OLS, and LightGBM does so by approximately 25%. The null of equal predictive accuracy, however, is not rejected for any of LightGBM's comparisons, so $H1_0$ cannot be rejected at conventional significance levels. The limited effective information in an eight-quarter test window and the dependence structure of panel forecast errors plausibly contribute to this result.

5.2. Valuation evidence

The Traditional RIV underestimates JPMorgan's market price at every valuation date, and the gap widens during 2025 while the model's inputs move only gradually. Replacing the persistence-based ROE projection with the LightGBM architecture, while leaving the remaining residual income architecture unchanged, reduces MAE by approximately 48%, RMSE by approximately 42% and MAPE by approximately 50%. With respect to H2, all three metrics are lower for the Enhanced model, which constitutes comparative empirical support for $H2_A$ within this application. This support is bounded: it concerns one institution over nine dates, measures proximity to observed market prices rather than to unobservable intrinsic value, reflects the complete architecture rather than the forecasting algorithm in isolation, and is not accompanied by a formal significance test.

5.3. Economic interpretation

The link between the two stages runs through a sequence of mechanisms. The forecasting model determines the ROE path; the ROE path determines the projected spread over the cost of equity; the spread, applied to book value under clean surplus, determines residual income; and discounted residual income, together with the persistence assumption and terminal value, determines intrinsic value. Each link introduces its own assumptions and sources of error. A better ROE forecast therefore does not guarantee a better valuation. In this application it coincides with one because the surrounding architecture—cost of equity, clean-surplus projection and terminal-value construction—is held constant and internally consistent.

Two features of the evidence qualify this interpretation. First, the forecasting model evaluated in Stage 1 (a one-quarter-ahead panel model) differs from the valuation-specific architecture used in Stage 2, and the Stage 2 benchmark is a persistence model rather than Pooled OLS; the relative magnitudes of the forecasting and valuation improvements are therefore not directly comparable. Second, book value accounts for most of intrinsic value (approximately 72% of the Traditional estimate at 2026Q1), so the improvement

concerns the component of value built on top of an already dominant accounting anchor rather than a distant terminal assumption. The persistent gap to market price in 2025 indicates that valuation accuracy also depends on inputs—the required return, the persistence assumption and market expectations beyond the explicit horizon—that an ROE forecasting model is not designed to capture.

The evidence does not establish that machine learning is superior to traditional valuation in general, that macroeconomic variables causally drive bank profitability, or that the Enhanced model recovers intrinsic value more accurately than the market. It establishes that, for the sample and period examined, a macro-informed machine-learning forecasting architecture produced lower ROE forecast errors in point-estimate terms and valuation estimates closer to observed prices than a fixed-persistence benchmark.

5.4. Contribution to the literature

The study contributes to the RIV literature by relaxing the fixed persistence assumption identified by [Dechow et al. \(1999\)](#) as a first-order driver of accuracy, replacing it with a data-driven near-term forecast combined with a capped long-horizon decay. Relative to [Liu and Hwang \(2025\)](#), who incorporate business-cycle information as a modifier of the price–accounting relation, the study uses macroeconomic and credit-cycle information as inputs to forecasting future profitability within the valuation model. It contributes to the machine-learning-in-finance literature, which concentrates on price prediction and classification ([Gu et al., 2020](#); [Petropoulos et al., 2020](#)), by evaluating continuous profitability forecasts within an accounting-based valuation framework. Methodologically, it separates forecasting performance from valuation usefulness and evaluates each with its own criteria and uncertainty, and it combines native and permutation importance to retain interpretability around a more complex forecasting model.

5.5. Practical implications

For equity analysts, the findings suggest that machine-learning profitability forecasts can supplement, rather than replace, a valuation grounded in accounting identities. A nonlinear forecast may provide a more responsive read on near-term profitability around turning points in the credit and monetary-policy cycle, but the statistical evidence counsels against mechanical reliance on any single algorithm’s point forecast. For bank management, the feature-importance evidence identifies profitability persistence, cost efficiency and credit quality as the inputs most informative for near-term performance in a data-driven model. For model-risk and governance functions, the models evaluated here are not validated for regulatory or supervisory use; the study instead illustrates a template in which a machine-learning component feeds a transparent, auditable accounting identity supported by documented validation checks.

6. Robustness, limitations and future research

6.1. Robustness and validation

Several design features address threats to validity. Look-ahead bias is controlled by a strict chronological protocol: in Stage 1, the scaler and all hyperparameters are fitted on the training window only; in Stage 2, the training sample at each valuation date contains only observations whose target quarter precedes that date, rolling features are lagged before any window opens, and training samples increase strictly across dates. A validation suite executed with each pipeline run confirms these properties and that the clean-surplus identity holds to floating-point precision at every step. Endogeneity in the panel models is mitigated by Fixed Effects, which removes time-invariant bank heterogeneity, and by the use of lagged rather than contemporaneous predictors. Driscoll–Kraay standard errors address the heteroskedasticity, serial correlation and cross-sectional dependence detected by diagnostic tests, and the Fixed Effects conclusions

are robust to dropping the most collinear predictor and to a comparison of coefficient signs with a Ridge regression. Overfitting risk is assessed through chronological out-of-sample performance, comparison of training and test errors, and re-estimation across random seeds: holding hyperparameters fixed across ten seeds, LightGBM's test performance has a coefficient of variation of 0.70%, the lowest among the four ensembles. This supports computational reproducibility conditional on the chosen specification and split; it does not show that the ranking or the negative generalisation gap would recur in a different test window.

6.2. Limitations

Several limitations qualify the findings. First, the forecasting sample comprises 20 large U.S. commercial banks under one regulatory regime, so results may not generalise to smaller or differently structured banks or to other jurisdictions. Second, the test window spans only eight quarters; together with cross-sectional and temporal dependence in forecast errors, this limits the power to confirm any model's superiority beyond point estimates. Third, both stages rely on a single historical episode—a training period followed by a test and valuation period spanning the transition from tightening to easing—so model rankings may depend on this episode. Fourth, machine-learning overfitting remains a risk, particularly in early expanding-window steps where training data are scarce. Fifth, the CAPM does not fully explain the cross-section of expected returns (Fama & French, 1992, 1993); because it is applied identically to both valuation models, its bias affects both, but it may contribute to the common gap to market price. Sixth, the terminal value divides quarter-20 residual income by the small quarterly cost of equity, which amplifies small estimation errors in residual income and the cost of equity; the persistence cap, acting through the quarter 9–20 path, mitigates but does not eliminate this sensitivity, which is shared by all perpetuity-based models. Seventh, the Enhanced architecture rests on progressively weaker empirical footing as the horizon lengthens: direct forecasts are the most directly validated component, the recursive phase inherits uncertainty from its forecast inputs, and the long-horizon phase remains assumption-dependent. Finally, the valuation evidence rests on a single firm and nine valuation dates, and should not be assumed to generalise to other institutions, sectors or periods.

6.3. Future research

The most immediate extension is to apply the valuation architecture to a cross-section of banks, which would establish whether the improvement documented for JPMorgan is a general property of embedding machine-learning forecasts in an RIV framework. Rolling-origin backtesting across multiple economic regimes would test whether the forecasting ranking, and LightGBM's negative generalisation gap, are stable properties of the data. Estimating SHAP attributions alongside permutation importance would extend the interpretation analysis. Explicit modelling of forecast and valuation uncertainty through prediction intervals would allow the statistical-power limitation to be represented directly in valuation output. Multi-factor cost-of-equity specifications and alternative terminal-value assumptions would clarify how much of the remaining gap to market price arises outside the profitability forecast.

7. Conclusion

This study examines whether macro-informed machine-learning forecasts of bank ROE improve residual income valuation. With respect to RQ1, LightGBM produces the strongest point-estimate forecasting performance among twelve econometric and machine-learning specifications (test RMSE 3.308; test R^2 0.405), and eight of nine machine-learning models improve on the Pooled OLS benchmark. Diebold–Mariano tests, however, do not establish statistically significant superiority of LightGBM over the main

benchmark models. With respect to RQ2, embedding a valuation-specific LightGBM architecture in a clean-surplus-consistent RIV framework reduces valuation error for JPMorgan Chase & Co. across nine valuation dates (MAPE from 44.07% to 21.91%; RMSE from \$121.25 to \$70.16) relative to a fixed-persistence benchmark, although a substantial gap to market price persists during 2025.

The study contributes a like-for-like forecasting benchmark, an auditable integration of machine-learning forecasts into accounting-based valuation, and an explicit separation of forecasting accuracy from valuation accuracy. The findings are specific to large U.S. commercial banks, a single test window, and one firm in the valuation stage, and should be read as evidence of improved market-price proximity rather than as a general demonstration of machine-learning superiority in equity valuation.

Declarations

Data availability. The empirical analysis uses quarterly bank-level financial statement and accounting data for U.S. commercial banks, supplemented with publicly available macroeconomic indicators from the [Federal Reserve Bank of St. Louis \(FRED\)](#) and equity risk-premium estimates from [Aswath Damodaran](#). Bank-level financial data were accessed through [LSEG Workspace](#) and include information derived from companies' reported financial statements, including balance-sheet, income-statement and related accounting variables.

The underlying company financial information is publicly disclosed, while the LSEG-delivered dataset and its structured distribution are subject to LSEG access and usage terms. Accordingly, the raw LSEG dataset is not redistributed with this study.

Code and research materials. The complete research code, model specifications, feature definitions, and supporting documentation are available at <https://github.com/Lv0733/ml-enhanced-bank-valuation>.

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Appendix A. Supplementary tables and figures

Table A1. Per-bank statistics for one-quarter-ahead ROE (%).

Bank	Ticker	Mean	Median	Std. dev.	Min	Max
Raymond James Financial Inc. ^a	RJF	16.25	17.02	2.28	11.28	19.24
East West Bancorp Inc.	EWBC	15.75	15.66	2.18	11.04	20.99
JPMorgan Chase & Co.	JPM	14.76	14.91	2.85	9.94	19.46
First Citizens BancShares Inc.	FCNCA	18.79	12.01	20.34	7.95	78.36
US Bancorp	USB	12.88	12.54	1.77	9.78	15.90
Fifth Third Bancorp	FITB	12.13	12.57	2.37	6.52	16.64
Wilson Bank Holding Co.	WBHC	12.18	11.97	1.32	10.51	14.82
UMB Financial Corp.	UMBF	11.08	11.14	2.52	7.08	15.03
First Horizon Corp.	FHN	11.18	11.03	2.71	6.31	20.55
Wintrust Financial Corp.	WTFC	10.78	11.08	1.63	7.38	13.95
Regions Financial Corp.	RF	10.85	10.84	3.03	4.18	16.23
Huntington Bancshares Inc.	HBAN	10.62	10.60	2.23	6.70	14.33
PNC Financial Services Group Inc.	PNC	10.29	10.61	2.30	4.34	14.17
M&T Bank Corp.	MTB	10.43	10.16	1.58	7.35	13.63
Webster Financial Corp.	WBS	10.33	10.42	1.99	4.95	13.60
KeyCorp	KEY	9.53	10.60	4.62	−2.20	16.02
Wells Fargo & Co.	WFC	9.88	10.85	3.14	1.06	12.76
Bank of America Corp.	BAC	9.77	9.73	1.51	6.73	12.38
Pinnacle Financial Partners Inc.	PNFP	8.91	9.40	1.50	6.20	11.30
Truist Financial Corp.	TFC	6.63	8.42	5.93	−12.67	11.31
Citizens Financial Group Inc.	CFG	7.47	7.43	1.61	4.60	10.48
Panel total		11.45	11.02	5.83	−12.67	78.36

^a Excluded from the Stage 1 estimation sample because of structurally unavailable credit-risk variables.

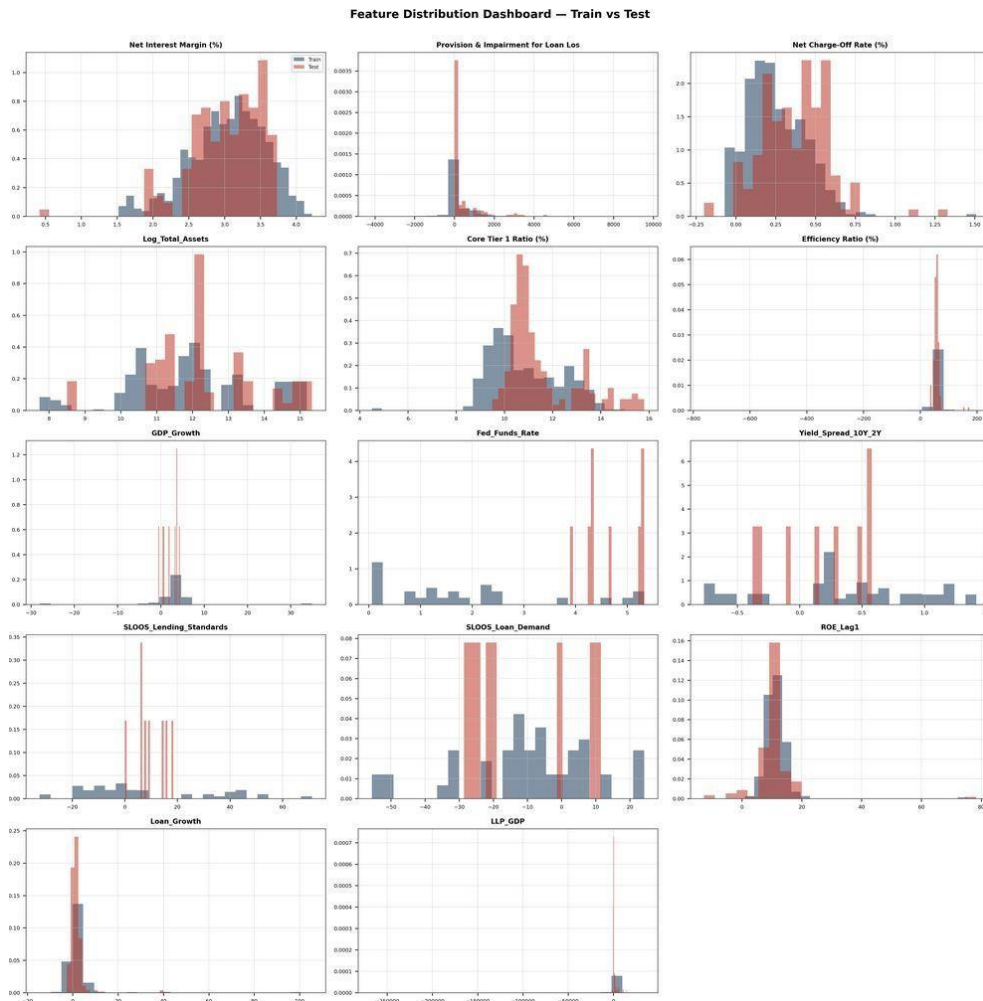


Figure A1. Distributions of the 14 numerical predictors in the training (2017Q1–2023Q4) and test (2024Q1–2025Q4) periods.

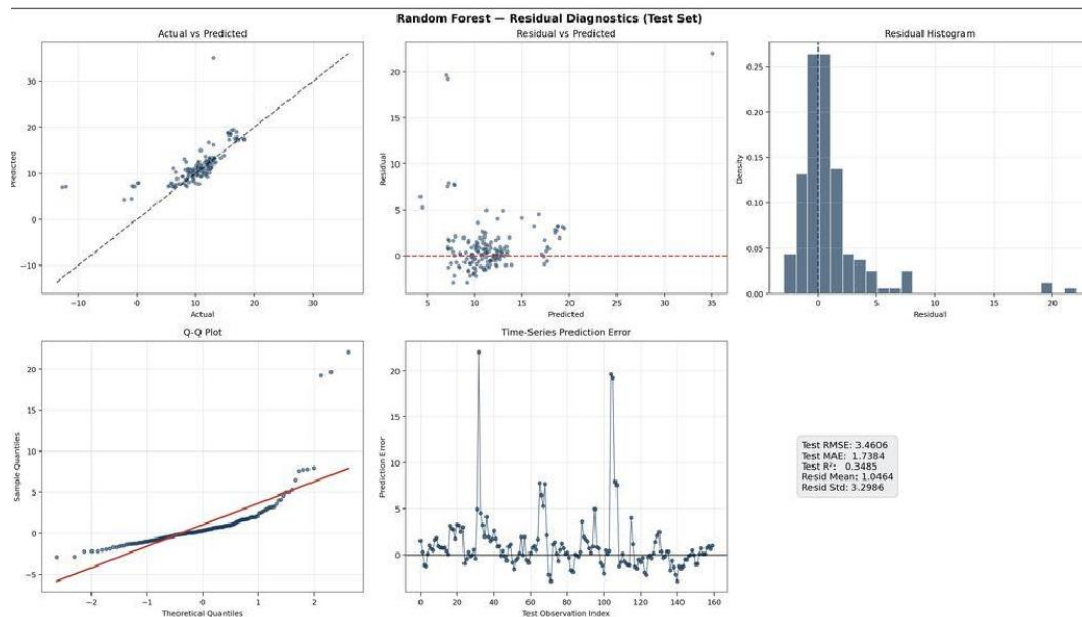


Figure A2. Residual diagnostics for Random Forest on the test set.

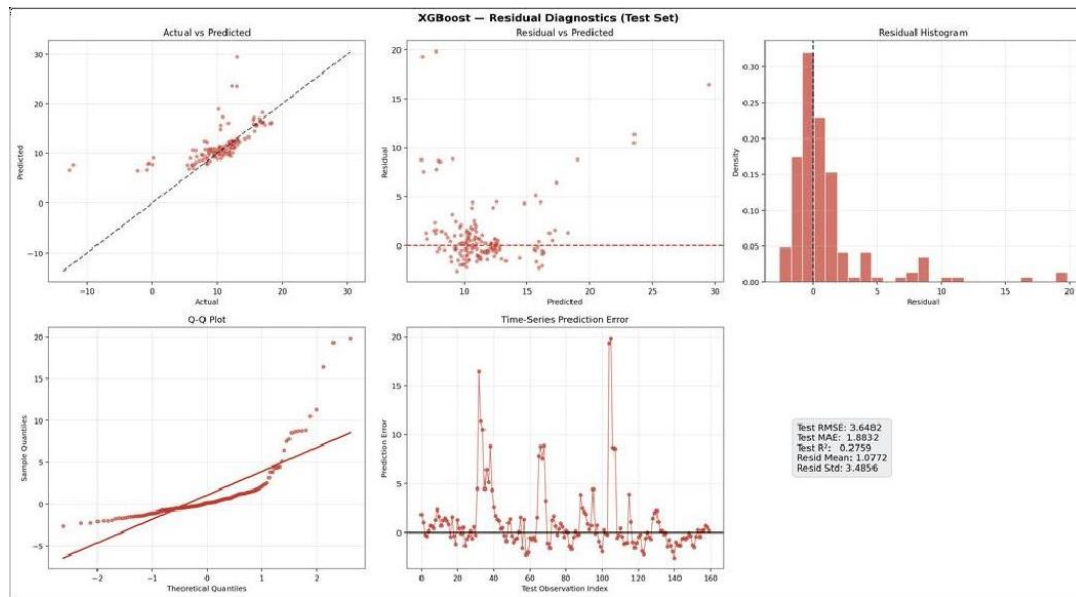


Figure A3. Residual diagnostics for XGBoost on the test set.

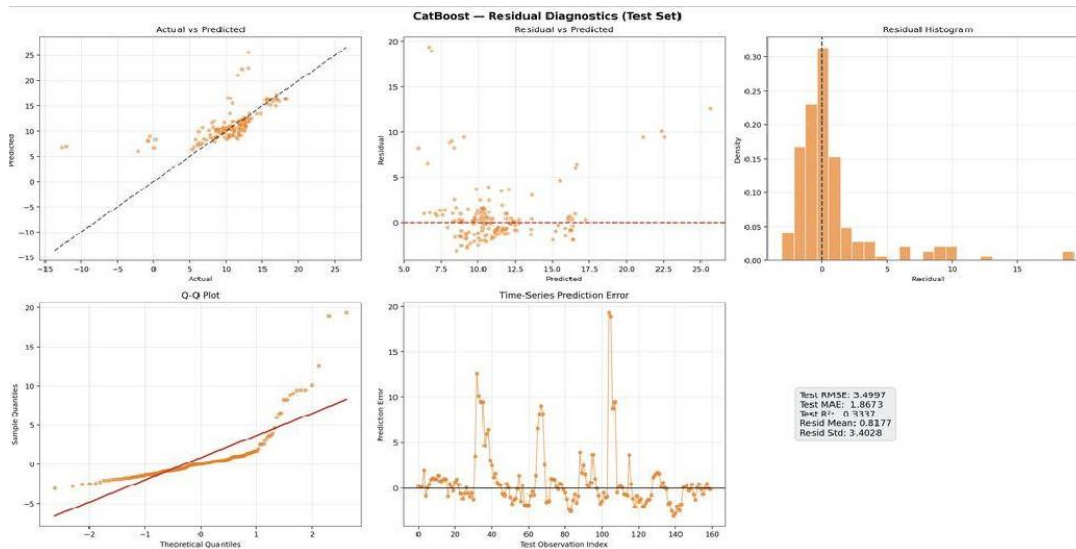


Figure A4. Residual diagnostics for CatBoost on the test set.